

Deterministic geometric Savitzky-Golay dynamics estimator

Shared parameter update: reference -> estimated

mass [kg]	2.351597591 -> 2.365808145	delta=+0.0142106
ratio=1.0060429		
CoG x [m]	-0.002024708562 -> -0.002358628816	delta=-0.00033392
ratio=-		
CoG y [m]	-3.052657894e-05 -> 0.0008683155458	delta=+0.000898842
ratio=-		
CoG z [m]	0.009509749599 -> 0.0105296429	delta=+0.00101989
ratio=-		
rotor effectiveness 1	1 -> 0.9653978103	delta=-0.0346022 ratio=0.96539781
rotor effectiveness 2	1 -> 0.9731335012	delta=-0.0268665 ratio=0.9731335
rotor effectiveness 3	1 -> 0.9580975087	delta=-0.0419025 ratio=0.95809751
rotor effectiveness 4	1 -> 0.954872828	delta=-0.0451272 ratio=0.95487283
rotor command lag [s]	0.004997730255 -> 0.1950022697	delta=+0.190005
ratio=-		

Inertia matrix [kg m^2]

reference:

```
[ 0.06500006148, -7.27899253e-07, 1.901508003e-05]
[-7.27899253e-07, 0.06495265634, 5.9167305e-08]
[ 1.901508003e-05, 5.9167305e-08, 0.1289921107]
```

estimated:

```
[ 0.06501097499, -2.504912063e-05, 2.255329729e-05]
[-2.504912063e-05, 0.06525605932, 7.259953625e-06]
[ 2.255329729e-05, 7.259953625e-06, 0.1289941558]
```

Principal moments: reference -> estimated

principal moment 1 [kg m^2]	0.06495264516 -> 0.06500843263	delta=+5.57875e-05
ratio=1.0008589		
principal moment 2 [kg m^2]	0.06500006701 -> 0.06525859291	delta=+0.000258526
ratio=1.0039773		
principal moment 3 [kg m^2]	0.1289921163 -> 0.1289941646	delta=+2.04828e-06
ratio=1.0000159		

Optimization

joint dynamics loss: 1.0830162382
Gaussian-prior cost: 0.337612667527
selected strict-ZOH rotor lag [s]: 0.195002269745

Bag failure_1_single

reconstruction position RMSE [m]: 0.000652642997862
reconstruction position component RMSE [m]: [0.00034 0.000324 0.000453]
reconstruction orientation RMSE [deg]: 0.596167087291
gyro consistency RMSE xyz [rad/s]: [0.041088 0.085974 0.026584]
specific-force consistency RMSE xyz [m/s^2]: [0.945152 0.162683 0.336919]
inferred external force RMS xyz [N]: [0.768267 0.386499 2.994559]
inferred external torque RMS xyz [N m]: [0.071397 0.249697 0.106052]
bag dynamics loss: 1.0830162382
gimbal command lag [s] 0.004998683929 -> 0.0363934178

Recorded command timestamp intervals (data-derived)

Bag failure_1_single

rotor: median dt=0.00499773026 s (200.091 Hz), min=2.00271606e-05 s, max=0.0138893127 s

gimbal: median dt=0.00499868393 s (200.053 Hz), min=4.529953e-06 s, max=2.48282003 s

Selected optimizer diagnostics

status=1 success=True nfev=4 njev=4
message='gtol' termination condition is satisfied.
reported optimality=1.412937660205671e-07
initial cost=1.4206294312964576
final cost=1.420628905726001
cost reduction=5.255704564888219e-07
coordinate step L2=0.00028345052051270204
final gradient inf-norm=1.412937660205671e-07
gradient tolerance satisfied=True
final Jacobian rank=14
final Jacobian cond(nonzero)=405.0893028901442
max directional finite-difference Jacobian relative error=7.33331540232405e-12